

Biomimetic snake robot

Mecanum chassis / robotic arm head / eight-servo tail

The project translates snake-like motion into phase-controlled servo joints. The result sits between animal gesture and a compact industrial machine.

This robot serves the 1001 Nights Traveler and the Snake Story

04

wheel motors

12

servo channels

08

tail joints



A mechanical snake

The design goal was not biological realism. We broke snake movement into controllable parts: a moving base, an expressive head, and a tail that carries a travelling wave.

Animal cue

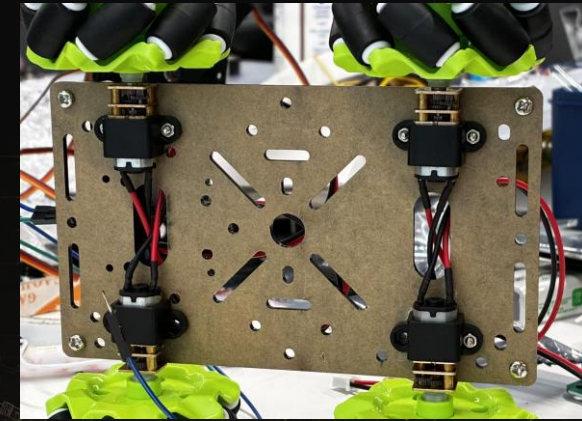
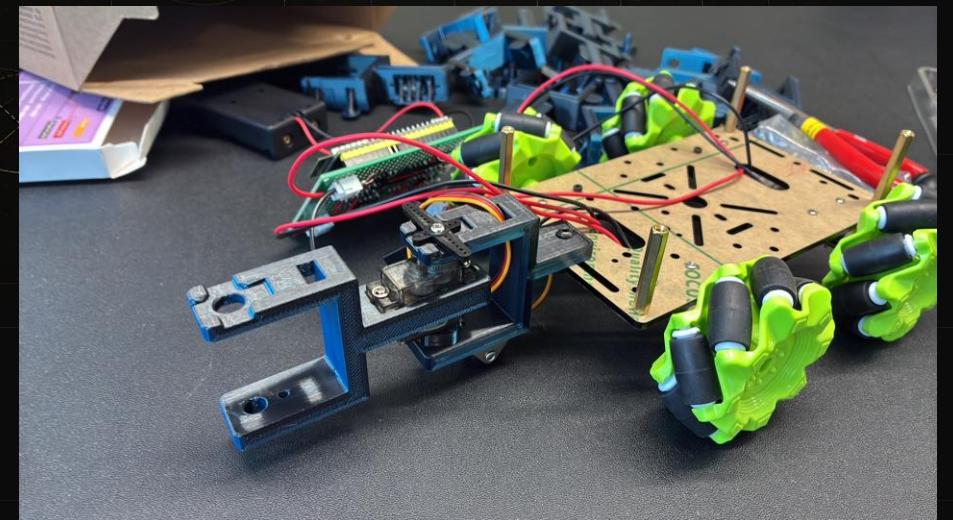
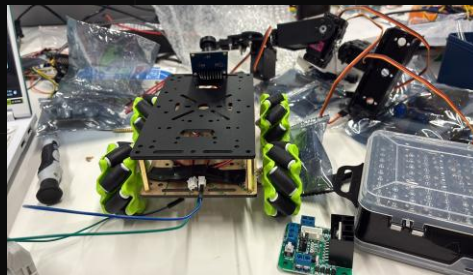
curved body, searching head, balancing tail

Motion rule

sine wave with a delay between joints

Machine body

four-wheel chassis and twelve servo outputs



We started with three separate systems

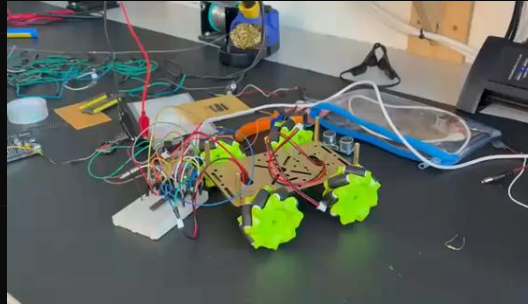
Arm, chassis, and tail were tested apart before we tried to make them behave as one robot.



01

Robotic arm head

Arduino and breadboard tests checked the basic servo control code before the head was mounted.



02

Mecanum chassis

A single SN754410 was tested first, because one chip can drive two DC motors. The layout was then scaled to four motors.



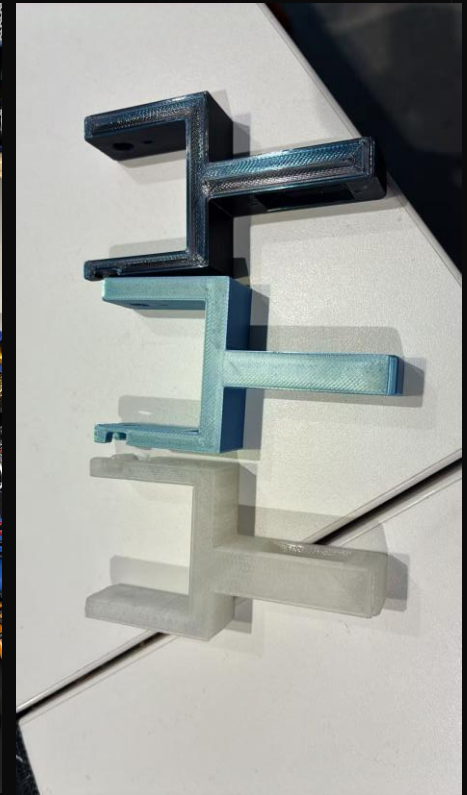
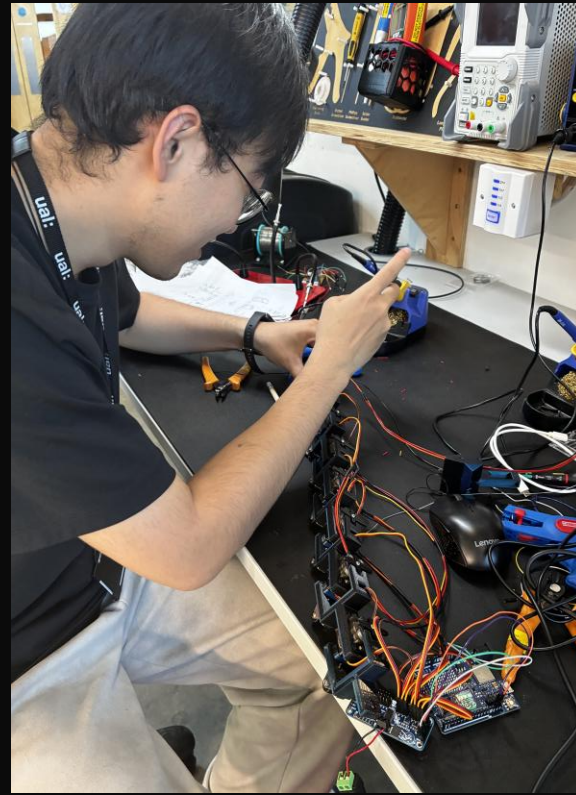
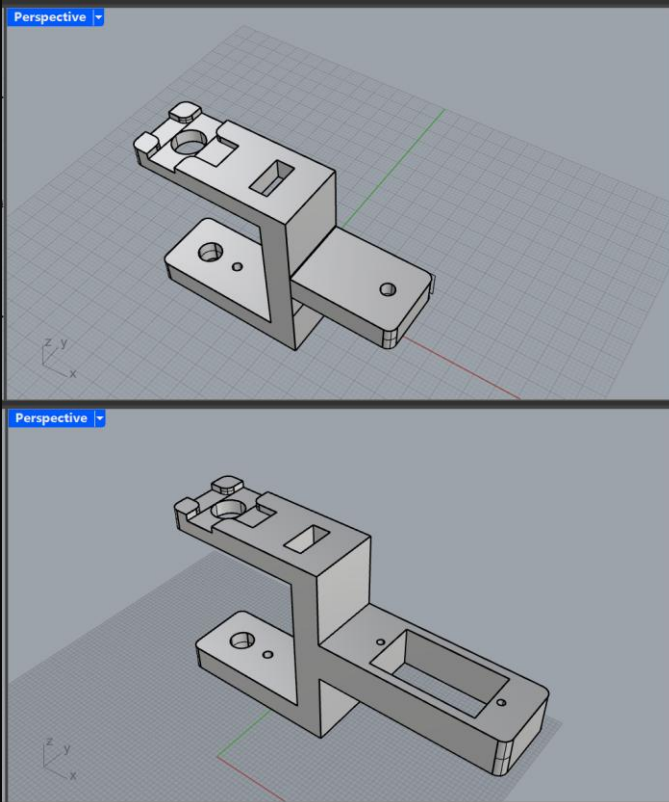
03

Eight-servo tail

The tail spacing was modelled around MF90 servo dimensions, then printed and assembled in sections.

The connectors connecting eight MF90 servo motors form a snake tail.

We printed multiple iterations before finally determining the appropriate dimensions and locking holes.



The cables can be organized more neatly.

Plan: Using the phase difference of sine waves to generate wave motion that travels along the body



CORE EQUATION

$$\theta_i(t) = A \cdot \sin(\omega t - i \cdot \Delta\phi) + \theta_{\text{offset}}$$

A

amplitude: left-right swing, for example $\pm 30^\circ$

ω

angular frequency: wave speed over time

i

joint index: 0 to 7 along the tail

$\Delta\phi$

phase shift: delay between neighbouring joints

θ_{offset}

centre position: usually 90°

```
const int numServos = 8;
float amplitude = 35.0, speed = 0.05, phaseShift = M_PI / 4.0, offsetAngle = 90.0;
unsigned long timeStep = 0;
unsigned long lastSnakeTime = 0;
```

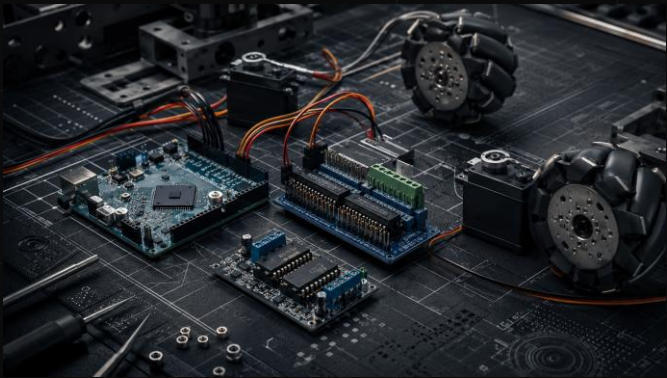
PHYSICAL LOCATION

First, I wrote code to move all the servos to 90 degrees, then fixed them to the 3D printed part and installed the servo arms.

```
for (int i = 0; i < 8; i++) {
  // 375 is typically a value with a pulse width of 1.5ms at 12-bit precision (0-4095)
  // representing the absolute neutral point of the servo motor.
  uint16_t pulse = map(90, 0, 180, 150, 600);
  pwm.setPWM(i, 0, pulse);
}
```

The chassis carries the biomimetic body

The chassis gives the robot reliable movement.



Arduino board
stable main control logic

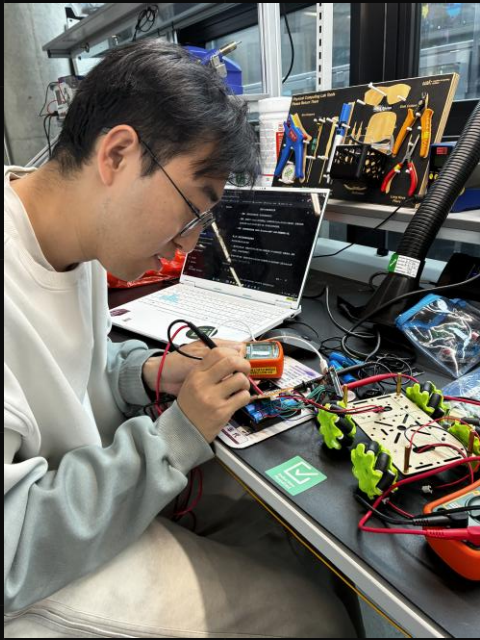
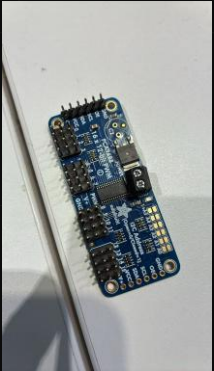
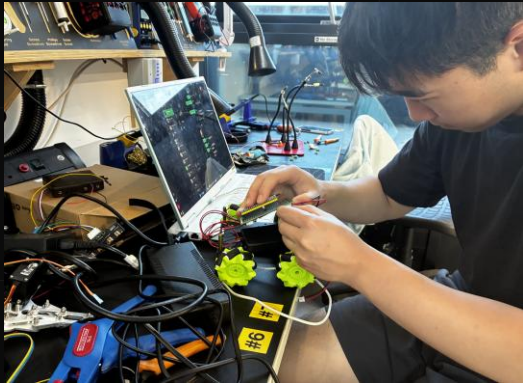
PCA9685
twelve servo PWM channels

3DOF arm head
four servos

Tail joints
MF90 × 8

SN754410 × 2
four DC motor outputs

Mecanum wheels
four driven wheels by N20 motor



01
stable Arduino
controller

12
PCA9685 servo
channels

04
independent wheel
drives

One board drives all twelve servos

PCA9685 let us treat the arm and tail as one PWM body. In code, each channel is just a joint angle.

CHANNEL MAP

0–3 robotic arm head / four servos

4–11 tail / eight MF90 servos

Once the mapping was fixed, the control problem became simpler: calculate twelve target angles, then send them to fixed channels.

Arduino

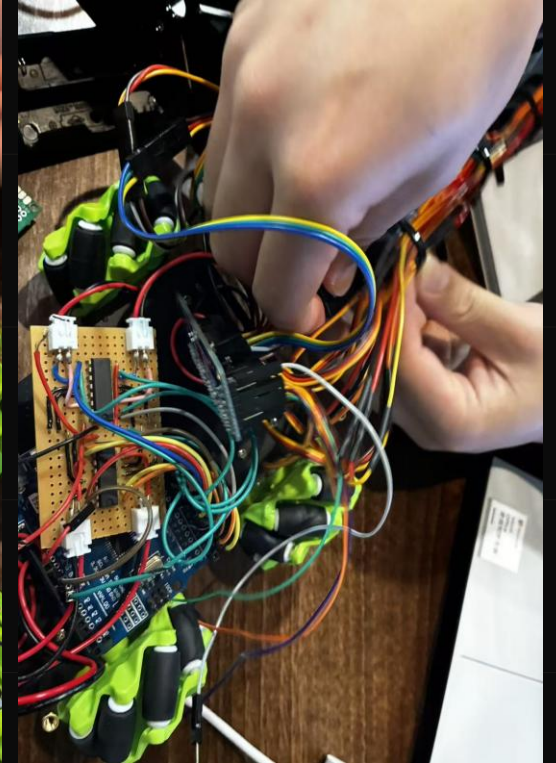
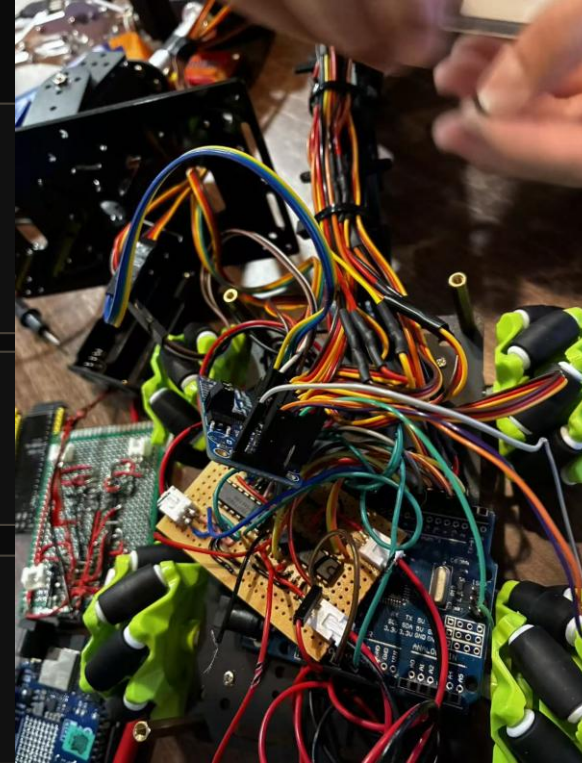
serial logic and I²C control

PCA9685

PWM timing board

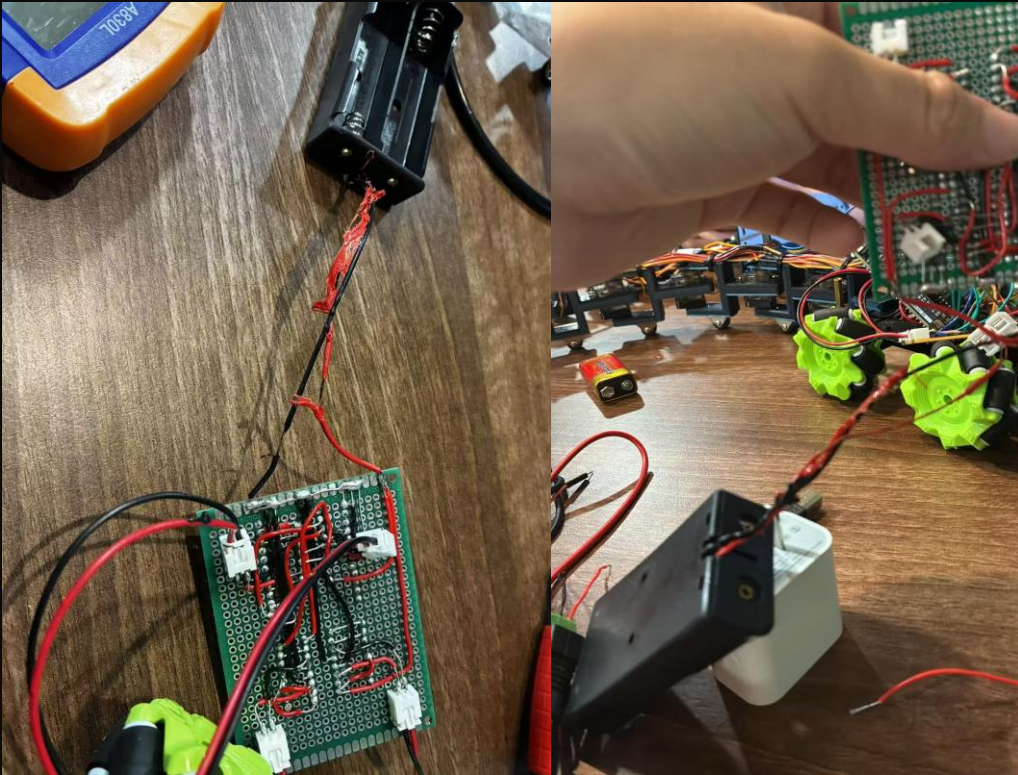
Servos [0...11]

head and tail joints



Failure set the final design limits

The electrical failures were not side notes. They defined what the robot could safely do in a final demonstration.



FAILURE TO DECISION

Low battery voltage

Motors and servos became unstable under load. Power had to be sized from the real current draw.

Short-circuit risk

The PCB reduced cable mess, but it made faults harder to isolate. Modular testing stayed necessary.

ESP32 damage

The board likely saw an abnormal current path or overloaded rail. Arduino was safer for the demo.

Servo current spike

Twelve servos can draw high peak current. Logic power and servo power should be planned separately.

Final rule: a system that repeats reliably is more valuable than a system with more pins but uncertain power behaviour.

The final prototype moves, bends, and gestures

The next version should move from open-loop motion to sensing, feedback, and repeatable behaviour.



OUTCOME

The build combines a four-wheel Mecanum base, a robotic arm head, an eight-servo tail, PCA9685 servo control, and a phase-shifted tail gait.

NEXT ITERATION

Add IMU or distance sensing for closed-loop posture.

Separate motor, servo, and logic power rails. Link tail gait with Mecanum direction control. Clean the wiring and make the body easier to repair.